

Experiment → Study of LAN Topology, Network devices and address Configuration.

Objectives

- ① To Study LAN Topology Used in College Campus
- ② To identify networking device and understand their function.
- ③ To analyze wired and wireless Communication, method used in network.
- ④ To measure speed of network for wired and wireless connection.
- ⑤ To obtain MAC, IP address and Subnet Mask.

Objective 1 → Study of College LAN Topology

→ Our College Campus network is built on a Hybrid Hierarchical Topology, meaning it combines trees, Extended Star, and ring layout to efficiently cover the grounds.
To manage the data traffic effectively

* The network is divided into three distinct layers.

- i) Core layer
- ii) Distribution layer
- iii) Access layer

① Core layer → Located in Central Server room
→ acts as the network backbone
→ It uses a ring topology to connect two L3 Switches for redundancy along side the main router and firewall.

② Distribution layer →

- This middle layer consists of four switches distributed across 1st, 2nd and 3rd floor of the main building.
- They serve as main distribution hub for each floor.

③ Access layer → The outermost layer contains three switches on the ground and third floors

- These provide the direct network connectivity to end-user devices, including lab system classroom ethernet ports, and wireless access points (APs)

Objective 2 → Networking devices and their function.

→ Based on our physical observation the Campus relies on few specific pieces of hardware to keep each layer functioning

- Router, Firewall and L3+ Switches:-

→ Housed in Core layer

→ The firewall secures the network, the router handles external traffic and the high capacity L3+ switches manages the heavy data flow coming from Campus Servers.

- TP-Link Omada SG3428XMP (L2+ Managed Switch)

→ Used primarily at the distribution layer

→ These 24-port switches allow network administrator to actively manage and monitor the traffic flow for each floor.

- TP-Link TL-SG1218MPE (Easy Smart Switch)

→ Access layer → it's generally used in access layer.

→ Crucially, 16 of its 18 ports support PoE+ (Power over ethernet)

→ This allows the Switch to send both data and electrical power through a single cable which is used to run Campus CCTU and Wireless APs.

- Wireless APs and Ethernet Ports:-

→ The final Connection points.

→ Ethernet wall ports provide direct, stable Connection for the lab Computers

→ While the APs broadcast the WiFi signal to student and faculty devices.

Objective 3 → Wired and wireless Communication Method.

→ The College network utilizes two primary method of Communication, to connect devices, to access layer.

- Wired Communication

→ This method used physical Ethernet Cable

→ Directly Connecting Stationary devices, like lab Computer System, into wall ports.

→ This provides a dedicated physical medium, for data to travel across different devices.

- Wireless Communication.

→ This method used Ceiling-mounted and outdoor wireless Access points

→ These Access points are used to transmit data using radio waves.

→ It acts as shared medium allowing students and faculty to connect multiple devices.

→ It doesn't require any kind of physical connection between sender and receiver.

Objective 4 → Speed and performance Comparison

- By observing the network behaviour.
- we noticed, distinct performance difference between the two connection method.

• Wired Network Method:-

→ Delivers much faster communication, and highly stable connection.

→ Because data travels over a dedicated physical cable,

→ There is very low latency, resulting in consistent overall performance for the devices connected with the ethernet cable directly.

- Wireless Network Method:-

- Generally exhibits slower communication
- It does provide a highly stable connection
- Because it relies on shared radio medium.
- Wireless connection has higher latency and its speed can negatively affected by physical objects.
- It provide slower speed compared to wired setup.
- It can also be interfered because of distance from AP and interference from having too many users connected at once

Objectives → Obtaining MAC, IP Address and Subnet Mask.

→ To test the network Configuration first hand. We used the windows Command prompt to retrieve the Connection details of a device on the Campus Network.

- Using ipconfig :- Running this Command displayed our Current IP v4 address

Eg → 10.38.232.xx

→ The default gateway and the Subnet mask which was.

Eg → 255.255.255.0

- Role of Subnet Mask →

→ It is crucial because it splits the IP address into two specific parts

- The network portion and the host portion
- This helps the system determine whether another device it wants to communicate with is on some local network or a completely different one.

• Using Getmac :-

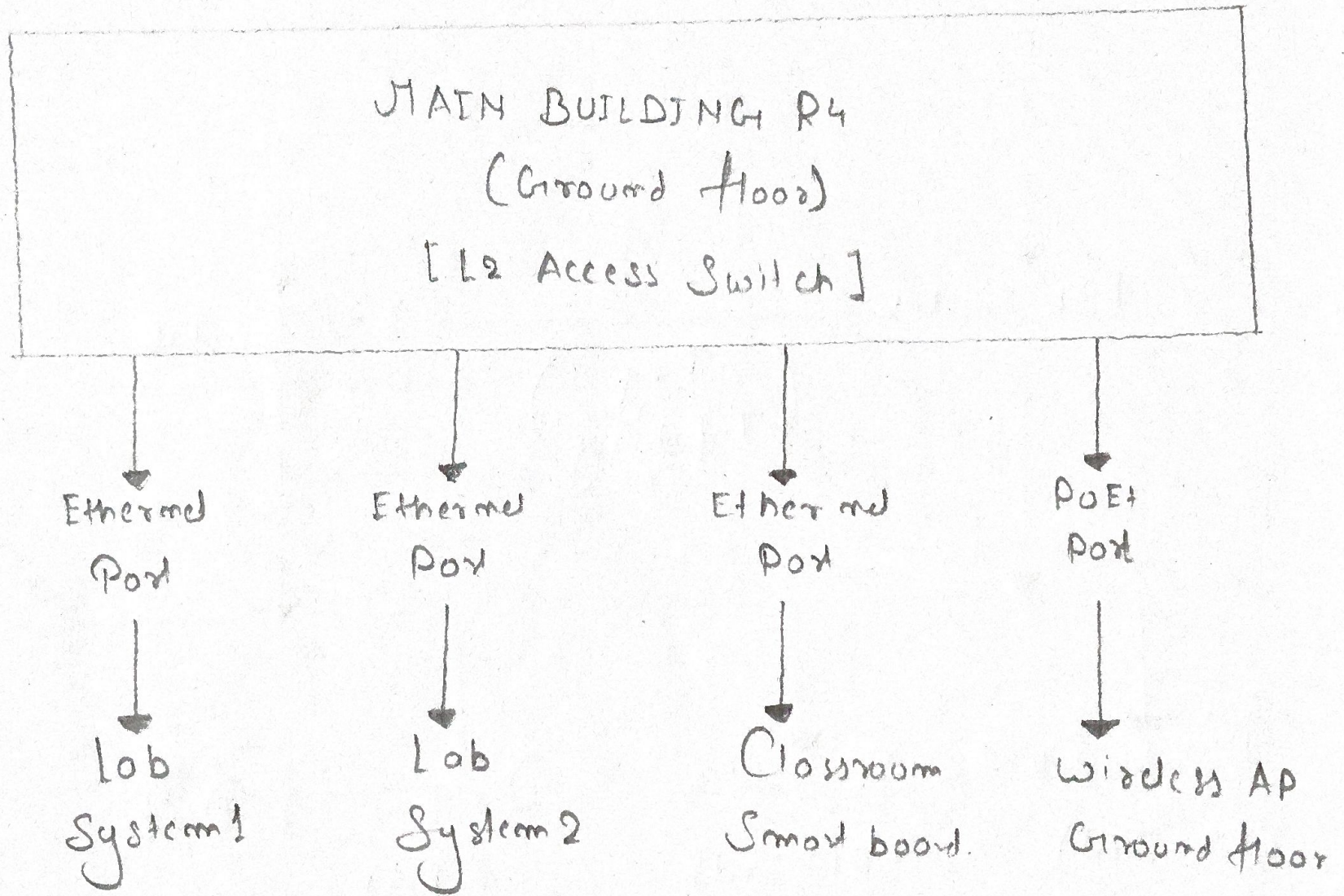
- Running this command allowed us to find physical address (MAC address)
- HardCoded directly into the computer network adapter hardware.

eg (34-74-96-xx-xx-xx)

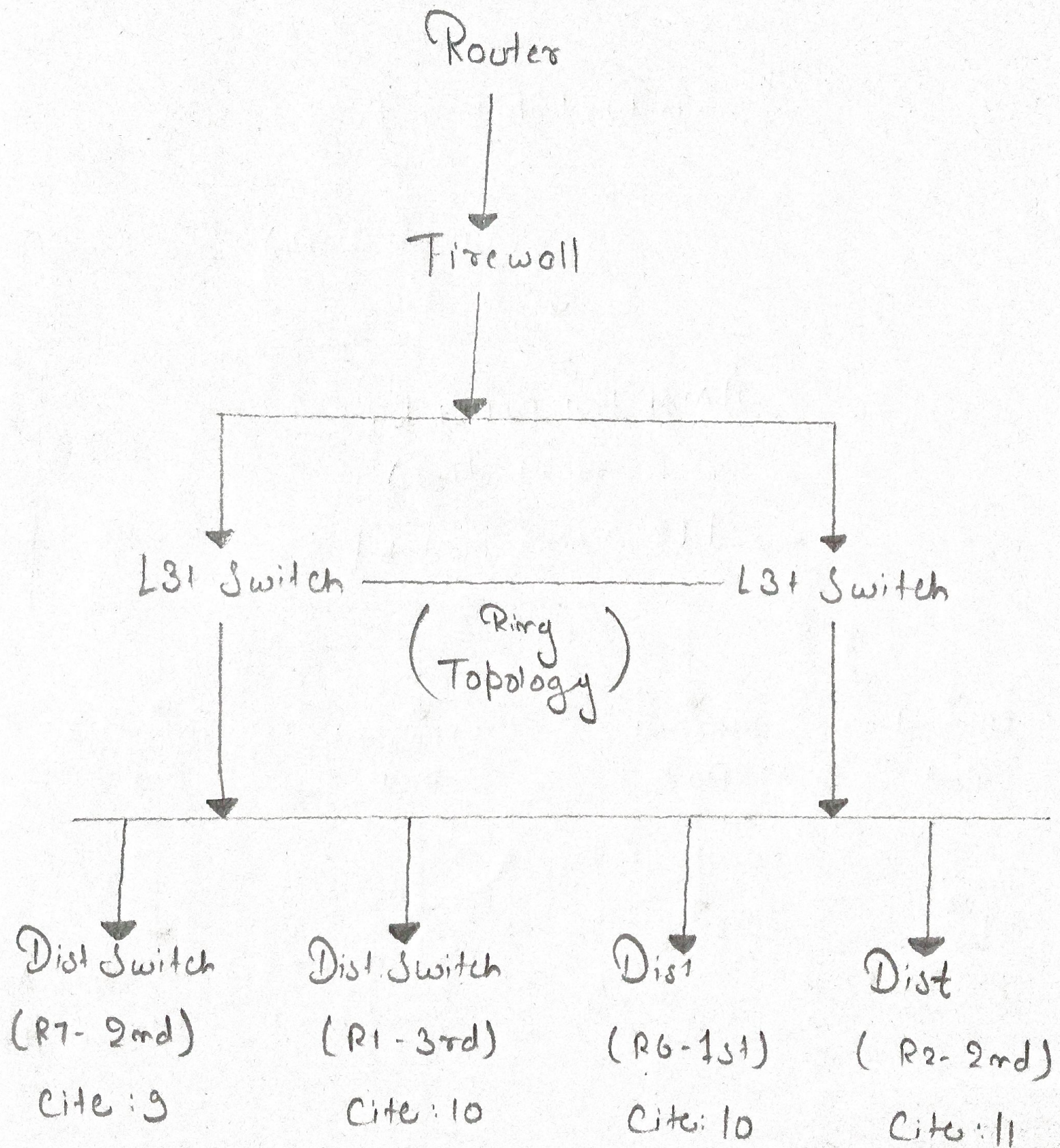
① The Ground Floor Access Layer

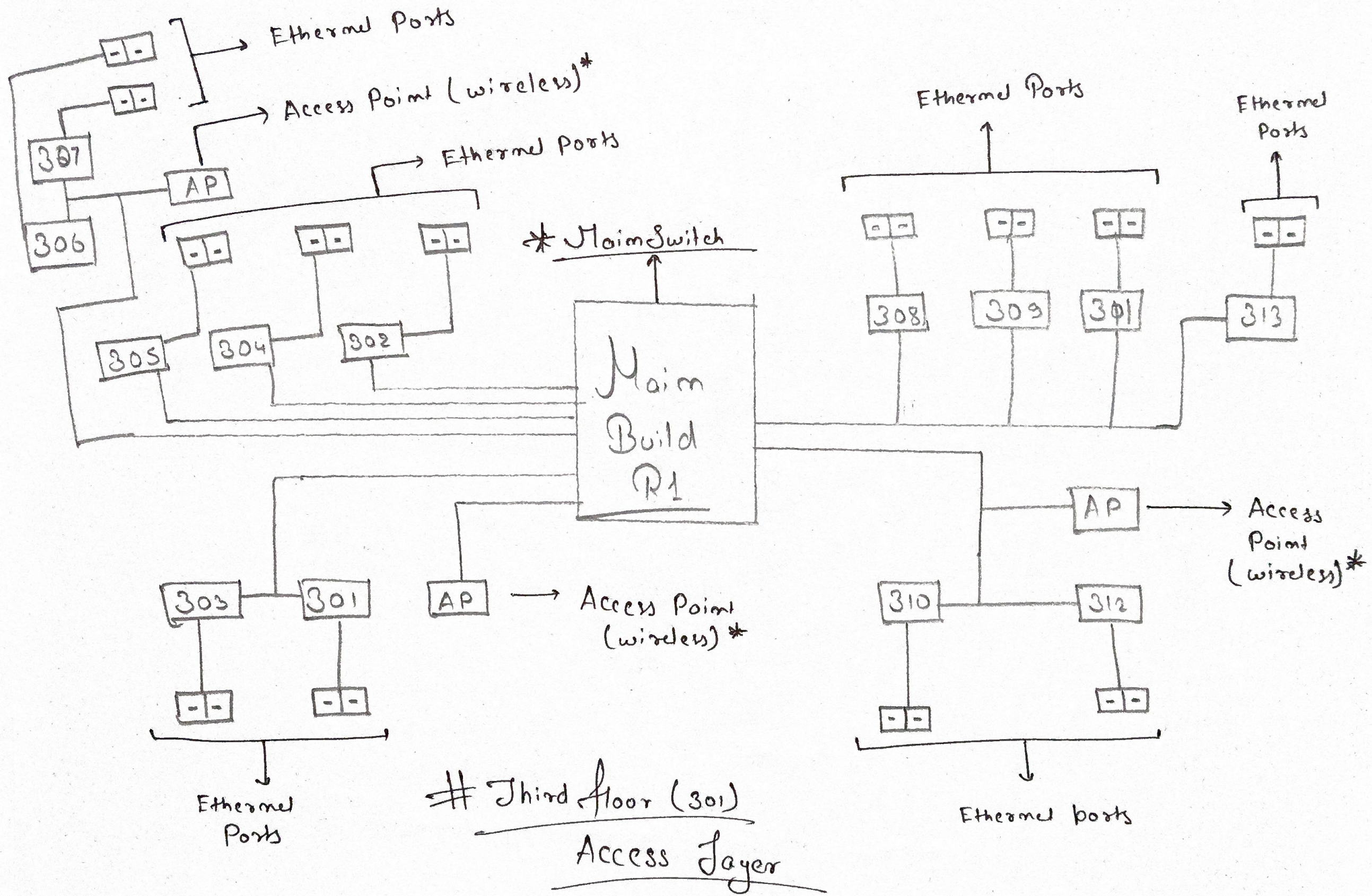
From Distribution Layer

(Uplink Cable)



② Core to Distribution Hierarchy (Server Room)





Difference between wired and wireless connection.

→ During the network test, we observed distinct performance difference between the two methods of communication.

- wired Network → Provides much faster communication and highly stable connection.

→ It has very low latency because it relies on dedicated physical medium. Using ethernet network cables to provide direct communication.

- wireless Network → Generally provide slower communication and higher latency compared to wired setups.

→ Instead of cables, it uses a shared radio medium to connect devices.

→ Because of this, the wireless connection's stability is easily affected by physical obstacles like distance from the router.

→ interference from having multiple users
connected at once.

Role of the Subnet Mask in the Network

→ When checking the network configuration.

Using the Command prompt, we reviewed
the Subnet mask. its role in the network is
crucial for few main reasons.

• IP Address Breakdown.

→ A Subnet mask's primary job is to take
an IP address and divide it in two parts,
the "network portion" and "host portion".

• Locating devices

→ By separating those two parts, it helps a
computer determine whether another device
it wants to talk to belongs in the same
local network or if it is located on a
completely different network.

- Network Efficiency :-

→ Ultimately, the Subnet mask is highly important for creating distinct network segment ensuring data routes efficiently and managing communication b/w device across the Campus.

Observation

Based on our practical study of College network infrastructure, we observed the following key points regarding how the system is managed.

- Centralized back bone :- The entire network relies on a centralized server room that acts as the primary backbone for all Campus communication.
- Redundancy and failover. Inside the server room, two L3+ Switches are connected in a redundant cycle structure.

- Efficiency Hardware:-

The network Cabinet across the diff. floor are equipped with merged.

TP link, PoE+ Switch Utilizing the allow for centralized management and provide a scalable architecture.

making it easy to expand.

Conclusion

In conclusion the observed College Campus LAN operates highly efficient.

Hybrid Hierarchical topology utilizing a mix of tree, Extended star and ring structure design. Ultimately this setup is highly scalable and perfect. balances the need of dedicated high speed wired lab connection with flexibility of Campus wide wireless access.